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**Key Points:**
- Collection of macrodispersivity from literature and evaluation of reliability
- Demonstration of inapplicability of an unique scaling law for macrodispersivity
- Emphasis on the need for aquifer characterization for transport modeling

**Supporting Information:**
- Supporting Information S1

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# Is unique scaling of aquifer macrodispersivity supported by field data?

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**Abstract** Spreading of conservative solutes in groundwater due to aquifer heterogeneity is quantified by the macrodispersivity, which was found to be scale dependent. It increases with travel distance, stabilizing eventually at a constant value. However, the question of its asymptotic behavior at very large scale is still a matter of debate. It was surmised in the literature that macrodispersivity scales up following a unique scaling law. Attempts to define such a law were made by fitting a regression line in the log-log representation of an ensemble of macrodispersivities from multiple experiments. The functional relationships differ among the authors, based on the choice of data. Our study revisits the data basis, used for inferring unique scaling, through a detailed analysis of literature marcodispersivities. In addition, values were collected from the most recent tracer tests reported in the literature. We specified a system of criteria for reliability and reevaluated the reliability of the reported values. The final collection of reliable estimates of macrodispersivity does not support a unique scaling law relationship. On the contrary, our results indicate, that the field data can be explained as a collection of macrodispersivities of aquifers with varying degree of heterogeneity where each exhibits its own constant asymptotic value. Our investigation concludes that transport, and particularly the macrodispersivity, is formation-specific, and that modeling of transport cannot be relegated to a unique scaling law. Instead, transport requires characterization of aquifer properties, e.g., spatial distribution of hydraulic conductivity, and the use of adequate models.

## 1. Introduction

Numerous laboratory column experiments of transport determined that a pulse injection of solute develops into a Gaussian bell curve. Its center travels with the fluid velocity $U$ and its longitudinal spread is quantified by a constant dispersion coefficient $D=\alpha_d U$, where $\alpha_d$ is the longitudinal pore-scale dispersivity. For granular media, $\alpha_d$ is constant for the sufficiently high *Peclet* numbers encountered in applications and is of order of the pore scale $d$, i.e., $\alpha_d \sim 10^{-2}-10^{-3}$ m.

For more general conditions of transport in a homogeneous and isotropic medium, the concentration of an inert solute $C(\vec{x},t)$ satisfies the advection-dispersion-equation (ADE) [*Bear*, 1972]

$$\frac{\partial C}{\partial t}+U\frac{\partial C}{\partial x_1}=D_l\frac{\partial^2 C}{\partial x_1^2}+D_{th}\frac{\partial^2 C}{\partial x_2^2}+D_{tv}\frac{\partial^2 C}{\partial x_3^2} \tag{1}$$

where $\vec{x}=(x_1,x_2,x_3)$ is the spatial coordinate vector and $t$ is the time. $D_l$, $D_{th}$ and $D_{tv}$ are the dispersion coefficients in longitudinal, transverse horizontal and transverse vertical direction. The flow velocity $U$ is assumed constant.

The transport behavior of a pulse of mass injected in the porous medium is commonly quantified by spatial moment analysis [*Dagan*, 1989]. The spatial plume movement is characterized by the first moment $L(t)$ of the concentration $C(\vec{x},t)$. The longitudinal spreading of the plume can be described by the second moment $S_{11}(t)$. The velocity and dispersivity are then given by

$$U=\frac{\mathrm{d}L(t)}{\mathrm{d}t}=\frac{1}{M_0}\frac{\mathrm{d}}{\mathrm{d}t}\int_{R^3}x_1C(\vec{x},t)\,\mathrm{d}\vec{x} \tag{2}$$

$$\alpha_d = \frac{1}{2U}\frac{\mathrm{d}S_{11}}{\mathrm{d}t} = \frac{1}{2UM_0}\frac{\mathrm{d}}{\mathrm{d}t}\int_{R^3}(x_1 - L_1(t))^2 C(\vec{x},t)\,\mathrm{d}\vec{x} \tag{3}$$

where $M_0$ is the mass of the injected pulse of solute. Similar definitions apply to transverse spreading but the present study addresses longitudinal spreading only and hereafter the term macrodispersivity refer to its longitudinal component.

In practice, transport parameters are often inferred from breakthrough curve (BTC) analysis [e.g., *Lenda and Zuber*, 1970]. The procedure is based on the characterization of the mass arrival at a specific position by using the distribution of travel time $\tau$. Similar relationships as (2) and (3) can be derived for the mean arrival time $T = \frac{x_1}{U}$ and dispersivity $\alpha_d = \frac{1}{2U^2}\frac{\mathrm{d}\sigma_\tau^2}{\mathrm{d}x_1}$, where $\sigma_\tau^2$ is the second central temporal moment. These relationships are used to derive the longitudinal dispersivity in practice even if the plume distribution is not Gaussian. For details on subsurface transport theory the reader is referred to a detailed discussion given in Appendix A1.

Numerous observations have revealed that the spreading of solutes at aquifer scale is characterized by much larger rates than those found in the small scale laboratory tests [e.g., *Pickens and Grisak*, 1981]. Assuming that the ADE (equation (1)) holds true, the relationships (equations (2) and (3)) were adapted to field scale by replacing the longitudinal pore-scale dispersivity $\alpha_d$ by the macrodispersivity $\alpha$, which is larger by orders of magnitude. This increase is attributed to the spatial variability of the hydraulic conductivity $K(\vec{x})$ of heterogeneous aquifers. Regarding $K(\vec{x})$ as random and relating flow and transport to the statistical properties of $K(\vec{x})$ has been one of the main objectives of the field of stochastic modeling, and extensive literature on the subject has been published in the last four decades [e.g., *Dagan*, 1989; *Gelhar*, 1993; *Rubin*, 2003]. A few fundamental issues were considered such as: How is $U$ related to the uniform mean head gradient $J$ for natural gradient flow? What is the scale of definition of $C$? Does the ADE with constant coefficients apply to field transport? If yes, how can $\alpha$ be related to the statistical moments of $K(\vec{x})$ and to local dispersion $\alpha_d$? What are the ergodicity requirements underlying the ADE with regard to the solute plume size? How can the ADE be extended for analysis of nonuniform flows, e.g., radial?

In parallel with the theoretical developments, continuing efforts have been invested in carrying out large scale transport experiments. In line with theoretical consideration, most of the experiments confirmed that macrodispersivity exhibits a scale dependence for early times of plume development. However, the preasymptotic scale-effect is aquifer specific, since the rate of growth with scale depends on the particular shape of the auto-correlation function and variance of the velocity distribution and thus on the aquifer structure. A detailed discussion on the scale effect from theoretical perspective is given in the Appendix A2.

The question of the asymptotic behavior of macrodispersivity at very large scale, i.e., in kilometre scale, is still a matter of debate. Most field studies are limited to travel distances of less than 100 m due to practical reasons. In addition, the asymptotic behavior might differ significantly between mildly and highly heterogeneous media.

One way to describe field-scale transport was to analyze collections of measured dispersivities from multiple aquifers. Values of $\alpha$ from individual aquifers were displayed graphically in dependence on the scale of observation $L$. This representation was motivated by the apparent increase of $\alpha$ with $L$ observed in a few experiments, and the models used in order to interpret them.

An early attempt to compile macrodispersivity data based on the available literature at the time was presented by *Arya et al.* [1988]. They distinguished between two clusters in the log-log graph: one for laboratory experiments and another one for field data. Based on the collection of single-aquifer macrodispersivities, *Arya et al.* [1988] defined an ensemble macrodispersivity $\hat{\alpha}$ by drawing a regression line and determining $\hat{\alpha}(L) = 0.229\ L^{0.755}$ (their equation (4a)), with $\hat{\alpha}$ and $L$ in meters. Though it is not stated explicitly, such an expression can be interpreted as an attempt to unify aquifers' behavior by a unique relationship.

*Neuman* [1990] took the approach a step further by considering a similar regression line to represent a unique scaling law of dispersivities. By using a larger database than the one employed by *Arya et al.* [1988], he arrives at a different regression line, of equation $\hat{\alpha}(L) = 0.0175\ L^{1.46}$ (his equation (22)). His $\hat{\alpha}(L)$ is stated to be valid over all scales, from the small laboratory ones to $L$ as large as $10^5$ m, with the interval of confidence of 95% represented in his Figure 1. The regression line fit of $\hat{\alpha}(L)$ to the data showed, that values at



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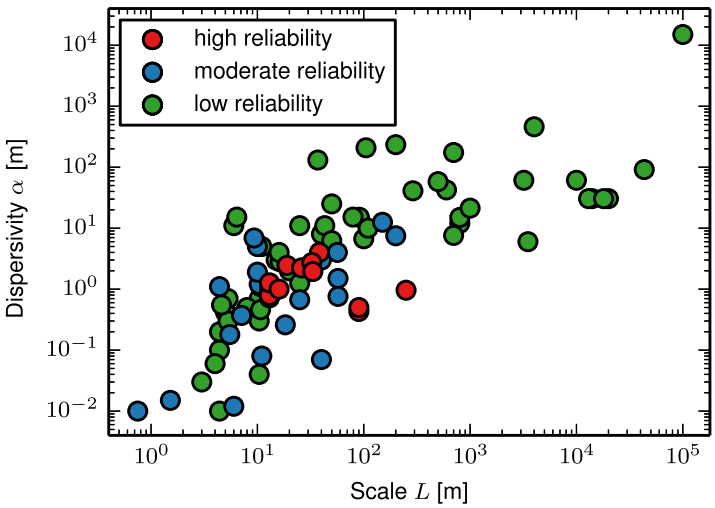


**Figure 1.** Longitudinal dispersivity versus scale of observation for experiments conducted in unconsolidated porous media with specification of reliability as reported by *Gelhar et al.* [1992].

distances larger than $L = 100$ m would much better fit an $\hat{\alpha}(L)$ with reduced slope. Therefore, *Neuman* [1990] suggests a modified equation of $\hat{\alpha}(L) = 0.32\ L^{0.83}$ for $L > 100$ m (his equation (32)), to account for the apparent reduction in the rate of increase with distance for large $L$.

A major step toward the elucidation of the scale dependence issue was achieved by the study of *Gelhar et al.* [1992] which expands on a previous report [*Gelhar et al.*, 1985]. First, the aquifer macrodispersivities $\alpha$ reported in the literature were summarized in a comprehensive manner (their Table 1). The authors listed detailed information about the various aquifers as well as flow and transport conditions. Secondly, *Gelhar et al.* [1992] carried out a critical analysis of the data which pointed out the difficulties in the interpretation of most reported values. The main result was the classification of the $\alpha$-data into three levels of reliability. The log-log plot of $\alpha$-data for unconsolidated media with different reliability symbols is reproduced herein as Figure 1. Though the data might indicate a scaling law for the ensemble macrodispersivity $\hat{\alpha}$, it is seen that most large values of $\alpha$ are of low reliability, while no such effect can be observed when considering only the highly and intermediate reliable ones. The authors warn against adopting a regression line: *. . .it does not seem reasonable to conclude that a single universal line [Neuman*, 1990*] can be meaningfully identified. [. . .] Rather we would expect a family of curves reflecting different dispersivities in aquifers with different degrees of heterogeneity*.

Despite the warning of *Gelhar et al.* [1992], further attempts has been made to identify a unique scaling law. *Xu and Eckstein* [1995] derived regression curves for the *Gelhar et al.* [1992] data by giving different weights to data of different reliability. They presented a functional relation between dispersivity and scale which is different from the one of previous studies. *Xu and Eckstein* [1995, Figure 1] display the *Gelhar et al.* [1992] data, the aforementioned regression lines of *Neuman* [1990] and the new ones $\hat{\alpha}(L) = 1.20\ (\log L)^{2.958}$ for equal weights and $\hat{\alpha}(L) = 0.83\ (\log L)^{2.414}$ for different weights (their equations (12b) and (14b)). A distinctive feature of their regression lines is that on the log-log paper the rate of change of $\log \hat{\alpha}(L)$ diminishes with $\log L$, almost toward a sill.

Subsequently, *Schulze-Makuch* [2005] added a few data points, representing additional field experiments. He proposed a regression line with the equation $\hat{\alpha}(L) = 0.085\ L^{0.81}$ for unconsolidated media (his Figure 1). His $\hat{\alpha}(L)$ is stated to be valid at all scales and irrespective of data reliability, with a 95% interval of confidence equal to 0.11. The discrepancy between this equation and the aforementioned ones of *Neuman* [1990] were the subject of an exchange between the authors [*Schulze-Makuch and Neuman*, 2006].

An apparently unique, or "universal" scaling law $\hat{\alpha}(L)$ suggested by a regression line through a collection of macrodispersivities $\alpha$ of different aquifers have prompted *Schulze-Makuch* [2005] to suggest in the abstract of the paper that *The data set presented here may provide ground water practitioners a preliminary guide to estimate dispersivity values at various scales and to guide and verify theories on scaling behavior*. Such a recommendation implies that the relationship $\hat{\alpha}(L)$ is capable to describe the macrodispersivity of any given aquifer at any scale, i.e., $\alpha \cong \hat{\alpha}(L)$, independent of aquifer properties such as variance and correlation structure or the flow configuration.

Indeed, it seems that practitioners use regression equations for predicting transport [*Fetter*, 2000]. In this context it is instructive to illustrate the prediction differences of the various relationships. Thus, the values of $\hat{\alpha}(L)$ for $L = 2000$ m based on *Arya et al.* [1988], *Neuman* [1990] for $L > 100$ m, *Xu and Eckstein* [1995]



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(with weights) and *Schulze-Makuch* [2005] are as follows: 71 m, 170 m, 15 m, and 40 m, respectively. Such significant differences in estimates of ensemble macrodispersivities point to the need to critically review the foundations of the universal scaling law.

The focus of this paper can therefore be summarized as: Is unique scaling of macrodispersivity supported by reliable field data? To answer that, we critically examine the literature database of single aquifer macrodispersivity and their interpretations. The renewed appraisal of macrodispersivity data is justified by the added perspective of stochastic modeling advances, new experimental field data and new characterization methods.

The concern here is primarily with the application of the universal scaling law $\hat{\alpha}(L)$ to estimate aquifer macrodispersivity $\alpha$, which we view as questionable for a few reasons: it may lead to overestimation of the rate of spreading; it is subject to large differences depending on the choice of the regression line; and it creates the belief that one can circumvent the thorough characterization of a particular aquifer and the specification of flow and transport conditions.

The discussion of the theoretical works or modeling approaches which were developed in relation with the unique scaling are beyond the scope of the present paper. Its focus is on the analysis of the field data which were used in order to justify the universal scaling.

The paper is organized as follows: In Sec. 2. we critically examine the field data. The criteria of reliability are updated. The studies listed by *Gelhar et al*. [1992] (being the database of most of the studies postulating a unique scaling law) were reevaluated leading to a exclusion of less reliable estimates of macrodispersivity. The remaining database of reliable macrodispersivities was complemented by reliable values from recent studies. In Sec. 3., we discuss the reevaluated field data by proposing a new representation of the experimental values. Along this line detailed field experiments in which $\alpha$ was measured at different distances $L$ are displayed. Sect. 4. summarizes and concludes the study. Detailed discussions of the various field experiments is provided in the supporting information to the article.

## 2. Reevaluation of Macrodispersivity Data From Literature

### 2.1. Field Studies Covered in *Gelhar et al*. [1992]

*Gelhar et al*. [1992] reviewed over 70 publications reporting macrodispersivity values. They compiled extensive tabulations of information on field site aquifer characteristics, experimental conditions and the method of data interpretation. They classified all values according to their reliability as high (R1), moderate (R2) and low (R3).

The evaluation by *Gelhar et al*. [1992] focused on experimental setting, test type, conservativeness of the tracer, the dimensionality of the monitoring as well as the type of analysis applied to the concentration data. Dispersivities of high reliability (R1) are thought to be accurate within a factor of two. Low reliability values (R3) are considered to be no more accurate than within one or two orders of magnitude. Intermediate reliabilities (R2) fall somewhere in between. The criteria for R1 and R3 as defined by *Gelhar et al*. [1992] are given in Table 1 (in italics).

The data set reported in *Gelhar et al*. [1992] contains 106 values of longitudinal dispersivity from experiments conducted at 59 field sites. Dispersivities were inferred from two types of subsurface solute transport events: controlled tracer experiments or uncontrolled tracer input at unknown scale, like contamination spills. The latter can either be naturally occurring events like tritium in groundwater from recharge containing atmospheric bomb tritium or human-induced events like leaks and spills from landfills, storage tanks and surface impoundments. Major problems in the interpretation of the uncontrolled events include unknown source input history and the sparse data networks relative to the extent of the plume. For controlled tracer experiments, often multiple observations were reported for one experiment, for instance at the Bonnaud site [*Molinari and Peaudecerf*, 1977; *Sauty*, 1977], or multiple experiments were performed at one site, for instance at Borden [*Sudicky et al*., 1983; *Sykes et al*., 1982a; *Egboka et al*., 1983; *Freyberg*, 1986; *Rajaram and Gelhar*, 1991]. For some experiments, multiple interpretations exist, e.g., at Mobile [*Papadopulos and Larson*, 1978; *Sykes et al*., 1982b].

About two-thirds of all experiments and contamination events took place in unconsolidated media (Figure 1), and these are the ones to be considered here. We limit our focus to dispersivities reported for unconsolidated media because transport in consolidated media often exhibits dual porosity behavior and transport along



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**Table 1.** Refined Criteria Employed for Assigning the Reliability of Macrodispersivity[a]

| Aspect | Criteria R1 | Criteria R2 | Criteria for Exclusion[b] |
|---|---|---|---|
| Test type | *regional flow with pulse/step input; radial diverging with pulse/step input; dipole test with pulse input* | radial converging with pulse input; dipole test with step input or recirculation | *single-well injection-withdrawal tests*; dipole test with little early time data |
| Source input history | *tracer input well defined* | source history thoroughly reconstructed | *unknown input quantity and duration*; spatially distributed sources |
| Tracer | *conservative* | nonconservative effects accounted for | *nonconservative effects not accounted for* |
| Observations | *3-D or 2-D (depth averaged over full aquifer thickness) monitoring*; high tracer mass recovery; ergodicity in plume development | monitoring over partial aquifer thickness; low tracer mass recovery | *measurement of tracer concentration in space was inadequate*: point sampling in vertical (individual depth) or horizontal (uncertain flow path) direction; low resolution of measurements |
| Method of moments | spatial or temporal moment analysis of Fickian tracer plume | adaption to nonuniform flow behavior | interpretation of bimodal curve with method of temporal moments |
| BTC analysis | BTC analysis with appropriate analytical solution (2-D, 3-D); adequate calibration (e.g., minimization of residuals) | BTC analysis with approximate solution; reduced dimensionality (1-D solution); simple calibration; type curve matching; pulse-width method; | *analysis not matched to flow configuration; BTC was assumed to be superposition of BTCs in separate hydro-stratigraphic units without data indication*; bimodal curve was interpreted using a single peak method; BTC analysis for individual depth |
| Numerical model | 2-D or 3-D model; fine grid resolution; adequate calibration | 1-D model; moderate grid resolution; simple calibration | coarse grid resolution; no calibration; very low sensitivity toward dispersivity; resolution of heterogeneous conductivity |

[a]The criteria defined by *Gelhar et al.* [1992] are embedded in italic. Details are given in sections 2.2 and 2.3 as well as in the supporting information section 1.
[b]Correspond to criteria of low reliability (R3) defined by *Gelhar et al.* [1992].

preferential flow paths. In those cases, transport cannot be assumed to be Fickian and alternative conceptualizations, like the the dual porosity model, may be required for transport description.

### 2.2. Revision of Reliability Criteria

In view of the advance in measurement techniques and methods of analysis, there is a compelling need to reevaluate the reported data, in order to assess the scaling behavior of macrodispersivity. Using the system established by *Gelhar et al*. [1992] as starting point, the criteria of reliability are expanded and refined with an emphasis on data analysis, listed in Table 1. For a study to be of a certain reliability (e.g., R2), all experimental aspects and the data interpretation fulfil at least the criteria of that class of reliability (e.g., R2 or higher). In the following, we shortly discuss the modifications compared to the systems of criteria defined by *Gelhar et al*. [1992].

The refined system given in Table 1 consists of three classes of reliability, with the major difference being that the criteria of low reliability (R3) are redefined as exclusion criteria. *Gelhar et al*. [1992] considered studies of low reliability (R3) to be no more accurate than within one or two orders of magnitude, which makes them highly questionable for further data interpretation like a regression analysis. Therefore, we surmise that data of this class of reliability shall be eliminated from the scale-dependence analysis unless their reliability can be upgraded; this issue is further detailed in section 2.3. The exclusion criteria in Table 1 include all the criteria of low reliability of *Gelhar et al*. [1992] (R3), with one change in the classification of dipole tests with step input or recirculation. Several criteria were added, in particular for the analysis of contamination events with numerical models. A detailed explanation of the exclusion criteria in combination with examples follows in section 2.3.

The criteria of high reliability are in line with those specified in *Gelhar et al*. [1992] (marked in italics in Table 1). For the class of moderate reliability (R2), *Gelhar et al*. [1992] did not specify criteria in detail. Explicit criteria for R2 were added: for the aspects test type, tracer and monitoring they are implicitly derived from *Gelhar et al*. [1992] as *somewhere between the extremes* of R1 and R3. The added criteria of a *thorough source reconstruction* allows contamination events to be of moderate reliability. Criteria according to the data analysis were added for both classes of reliability R1 and R2, with differentiation between statistical, analytical and numerical data interpretation. This has not been done by *Gelhar et al*. [1992] for R1 and R2; however, we think that this is necessary with regard to the evaluation of newly published studies. We do not discuss the differences between R1 and R2 in detail because studies of both classes are of sufficient reliability to be taken into account for the scale-dependence analysis (section 3). Instead, the reader is referred to the specification of the refined criteria in section 1 of the supporting information.



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**Table 2.** Reevaluated Macrodispersivities $\alpha$ at Maximal Observation Scale $L$ of High ($R = 1$) and Medium ($R = 2$) Reliability From Literature[a]

| Site Name[b], Country | Author(s) | Source[c] | $L$ (m) | $\alpha$ (m) | R |
|---|---|---|---|---|---|
| Bonchon Study Area, Korea | *Kim et al.* [2005] | new | 4.5 | 0.67 | 2 |
| Bonnaud (1), France | *Molinari and Peaudecerf* [1977]; *Sauty* [1977] | G92 | 32.5 | 2.7 | 1 |
| Bonnaud (2), France | *Sauty et al.* [1978] | G92 | 13 | 1 | 2 |
| Borden (1), Canada | *Freyberg* [1986]; *Rajaram and Gelhar* [1991] | G92 | 90 | 0.5 | 1 |
| Borden (2), Canada | *Rivett et al.* [1994] | SM05 | 45 | 0.5[d] | 1[e] |
| Borden (3), Canada | *Sykes et al.* [1982a] | G92 | 700 | 3.1[d] | 2[e] |
| Borden (4), Canada | *Sudicky et al.* [1983]; *Gelhar et al.* [1985] | G92 | 11 | 0.34 | 2[f] |
| Burdekin Delta, Australia | *Wiebenga et al.* [1967]; *Lenda and Zuber* [1970] | G92 | 18.3 | 0.26 | 2 |
| Burnham Aquifer, New Zealand | *Pang and Close* [1999] | new | 90 | 5.3 | 2 |
| Byles-Saint Vulbas (1), France | *Rousselot et al.* [1977] | G92 | 7.1 | 0.37 | 2 |
| Cambridge site, Canada | *Robertson et al.* [1991] | SM05 | 130 | 1 | 2[e] |
| Campuget, France | *Iris* [1980] | G92 | 40 | 3 | 2 |
| Cape Cod, Massachusetts, USA | *Garabedian et al.* [1991] | G92 | 212[d] | 0.96 | 1 |
| Chalk River Site (1)/Twin Lake Aquifer, Canada | *Moltyaner et al.* [1993] | SM05 | 266 | 0.55 | 1 |
| Chalk River Site (2), Canada | *Pickens and Grisak* [1981]; *Huyakorn et al.* [1986b] | G92 | 8 | 0.5 | 2[e] |
| Corbas, France | *Sauty* [1977]; *Welty and Gelhar* [1989] | G92 | 150 | 10.5 | 2[f] |
| Dornach, Germany | *Klotz et al.* [1980] | G92 | 10 | 1.9 | 2 |
| Gas Plant Facility, Michigan, USA | *Chiang et al.* [1989] | SM05 | 350 | 0.8[d] | 2[e] |
| Grenoble Aquifer, France | *Courtois et al.* [2000] | new | 45 | 7 | 2 |
| Hanford (1), Washington, USA | *Bierschenk* [1959]; *Cole* [1974] | G92 | 3500 | 6 | 2[e] |
| Hebei Province Aquifer, China | *Yang et al.* [2001] | new | 15.5 | 1.72 | 2 |
| Heretaunga aquifer (1), Roys Hill Site, New Zealand | *Thorpe and Barry* [1977] | G92 | 25 | 0.67 | 2 |
| Heretaunga aquifer (2), Flaxmere Site, New Zealand | *Thorpe and Barry* [1977] | G92 | 55–59 | 1.4–11.5 | 2 |
| Horkheimer Insel (1), Germany | *Ptak and Teutsch* [1994] | SM05 | 56.4 | 6 | 2 |
| Horkheimer Insel (2), Germany | *Ptak and Teutsch* [1994] | SM05 | 17.1 | 2.5 | 2 |
| Jutland Aquifer, Denmark | *Jensen et al.* [1993] | SM05 | 200[d] | 0.45 | 1[e] |
| Kesterson Aquifer, California, USA | *Hyndman and Gorelick* [1996] (Reanalysis) | SM05 | 13.8[d] | 1.06[d] | 2[f] |
| Koeberg Nuclear Power Station, South Africa | *Meyer et al.* [1981] | G92 | 8[d] | 0.42 | 2[e] |
| Krauthausen, Germany | *Vereecken et al.* [2000] | new | 170 | 3.64 | 1 |
| Lauswiesen Site, Germany | *Händel and Dietrich* [2012] | new | 52 | 6.25 | 2 |
| Lower Glatt Valley (1), Switzerland | *Hoehn & Santchi* [1987] | G92 | 110 | 10 | 2[e] |
| MADE, Mississippi, USA | *Adams and Gelhar* [1992] | G92 | 200 | 7.5 | 2 |
| Meredosia, Illinois, USA | *Naymik and Barcelona* [1981] | G92 | 164[d] | 2.13–3.35 | 2[e] |
| Mobile (1), Alabama, USA | *Huyakorn et al.* [1986b]; *Molz et al.* [1986] | G92 | 38.3 | 4 | 1 |
| Mobile (2), Alabama, USA | *Sykes et al.* [1982b]; *Papadopulos and Larson* [1978] | G92 | 57.3 | 0.76 | 2 |
| New Mexico State University, New Mexico, USA | *Kies* [1981] | G92 | 25 | 1.6 | 2[e] |
| Palo Alto, California, USA | *Valocchi et al.* [1981]; *Roberts et al.* [1981] | G92 | 16 | 1 | 1 |
| Polish Aquifer, Poland | *Kreft et al.* [1974] | G92 | 5–6 | 0.18 | 2 |
| Rabis Creek Catchment, Denmark | *Engesgaard et al.* [1996] | SM05 | 1000 | 1–10 | 2 |
| Rocky Mountain Arsenal (1), Colorado, USA | *Thorbjarnarson and Mackay* [1997] | new | 5.2 | 1.2 | 2 |
| Scarborough Aquifer, Canada | *Palmer and Nadon* [1986] (Reanalysis) | SM05 | 6.5 | 0.84[d] | 2[f] |
| Stanton (1), Texas, USA | *Broermann et al.* [1997] | new | 15 | 3.78 | 2 |
| Testfeld Sued, Germany | *Bösel et al.* [2000] | SM05 | 80 | 5 | 2 |
| Torcy, France | *Dieulin* [1980] | G92 | 15 | 2.14[d] | 2[e] |
| Tucson, Arizona, USA | *Wilson* [1971]; *Welty and Gelhar* [1989] | G92 | 79.3 | 1.2 | 2[f] |
| UC Berkeley, California, USA | *Lau et al.* [1957] | G92 | 19 | 2.14[d] | 1 |
| Zeitz, Germany | *Gödeke et al.* [2006] | new | 55 | 0.6 | 2 |

[a]Details on every study are given in the supporting information.
[b]Number in brackets for sites with multiple experiments.
[c]*G92—Gelhar et al.* [1992], *SM05—Schulze-Makuch* [2005], *new—newly evaluated*.
[d]Modification of $L$ or $\alpha$ compared to reference source.
[e]Modification of reliability compared to reference source.
[f]Modification of reliability after reinterpretation of experiment.

Our aim is to extract dispersivities from literature which are of sufficient reliability to serve further analysis. The assignment of reliability classes or exclusion criteria is not intended to pass judgement on the worth of the underlying study, but rather to provide tools for assessing the quality of the estimation of longitudinal dispersivity. Many of the reported experiments were not conducted with the primary objective to characterize longitudinal dispersivity with the highest possible accuracy.

### 2.3. Reevaluation of Gelhar's Data

We have embarked upon the detailed examination of all experiments in unconsolidated media reported by *Gelhar et al.* [1992] by expanding and refining reliability criteria (Table 1). An elaborate description of the reevaluation of the studies is given in the supporting information. The results are summarized in Tables 2



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**Table 3.** Excluded Dispersivities $\alpha$ at Maximal Observation Scale $L$ From Gelhar et al. [1992] With Short Statement of the Reason of Exclusion[a]

| Site Name[b], Country | Author(s) | $L$ (m) | $\alpha$ (m) | Reason |
|---|---|---|---|---|
| Aefligen, Switzerland | *Werner* [1983] | 200 | 234 | tracer[c] |
| Aquia Formation, Maryland, USA | *Wood* [1981] | 100000 | 5600–40000 | tracer[c] |
| Arkansas River Valley, Colorado, USA | *Konikow and Bredehoeft* [1974] | 18000 | 30.5 | source[d] |
| Barstow (1), California, USA | *Robson* [1974] | 6.4 | 15.2 | test type[e] |
| Barstow (2), California, USA | *Robson* [1974, 1978] | 10000/3200 | 61 | source[d] |
| Bonsall Subbasin, California, USA | *Helweg and Labadie* [1977] | 14000 | 30.5 | source[d] |
| Borden (5), Canada | *Egboka et al.* [1983] | 600 | 30–60 | source[d] |
| Brook Fields, Amherst, Massachusetts, USA | *Leland and Hillel* [1982] | 4 | 0.05–0.07 | local[f] |
| Byles-Saint Vulbas (2), France | *Rousselot et al.* [1977]; *Sauty* [1977] | 10.7 | 0.46/1.1 | data[g] |
| Byles-Saint Vulbas (3), France | *Rousselot et al.* [1977]; *Sauty* [1977] | 9.3 | 6.9 | data[g] |
| Cambric Test Site, Nevada, USA | *Daniels* [1982]; *Thompson* [1988] | 91 | 15.1[h] | source[d] |
| Chalk River Site (3), Canada | *Pickens and Grisak* [1981] | 3 | 0.03 | test type[e] |
| Chalk River Site (4)/Perch Lake, Canada | *Lee et al.* [1980] | 4[h] | 0.012 | local[f] |
| Chalk River Site (5)/Twin Lake Aquifer, Canada | *Moltyaner and Killey* [1988] | 40 | 0.07[h] | local[f] |
| Hanford (2), Washington, USA | *Ahlstrom et al.* [1977] | 20000 | 30.5 | source[d] |
| Heretaunga aquifer (3), Rubbish dump, New Zealand | *Thorpe and Barry* [1977] | 290 | 41 | source[d] |
| Land fill near Lyon, France | *Fried* [1975] | 600–1000 | 12 | source[d] |
| Lower Glatt Valley (2), Switzerland | *Hoehn* [1983]; *Hoehn and Santschi* [1987] | 10.4 | 0.04/0.3/0.7 | local[f] |
| Rhine Aquifer near Mulhouse, France | *Fried* [1975] | 600[h] | 15 | source[d] |
| Rhine Aquifer near Strasbourg, France | *Fried and Ungemach* [1971] | 6 | 11 | data[g] |
| Rocky Mountain Arsenal (2), Colorado, USA | *Konikow* [1976] | 9000[h] | 30.5 | source[d] |
| South Farmingdale, Long Island, New York, USA | *Pinder* [1973] | 1300[h] | 21.3 | source[d] |
| Spokane River Site, Washington Idaho, USA | *Vaccaro and Bolke* [1983] | 43400 | 91.4 | source[d] |
| Stanton (2), Texas, USA | *Wood and Ehrlich* [1978] | 1.5 | 0.015 | local[f] |

[a]Detailed discussion of every study is given in the supporting information.
[b]Number in brackets for sites with multiple experiments.
[c]Nonconservative effects of tracer not accounted for.
[d]Source input history unknown.
[e]Test type of low reliability.
[f]Local dispersivity.
[g]Inadequate measurements, low reliability of data interpretation.
[h]Modification of $L$ or $\alpha$ compared to reference source.

and 3: studies of R1 and R2, including those upgraded from the R3 class as well as excluded studies, belonging primarily to the eliminated R3 class with a short note on the reason for exclusion.

In view of the interpretations of a single experiment at different times, the reevaluation includes a reduction of the data set to one analysis per experiment. This can be one or several dispersivity values if an evolution of long-range dispersivity was reported. We do not include multiple values of dispersivity at the same distance, which resulted from repetitions of experiments or an analysis for different hydraulic units of the same aquifer. However, different experiments at one field site, as at the Borden site, are included, if they are of sufficient reliability.

*Gelhar et al*. [1992] evaluated dispersivities of 6 experiments in unconsolidated media as having reliability R1, which we keep unchanged. Out of the 18 original R2 experiments in unconsolidated media, 10 are retained, 3 have alternative interpretations which we use instead and 5 are excluded. From the 35 experiments of low reliability, we reevaluate 8 as having a moderate reliability R2; the remaining are excluded. In the following, we give a brief discussion of the procedure leading to the exclusion criteria, summarized in the Table 1. All experiments are discussed in detail in the supporting information.

Several criteria are assessed before making the decision to exclude. The first criterion for exclusion is the test configuration. We exclude dispersivities determined from single-well injection-withdrawal tests, which essentially determine local dispersivity [*Pickens and Grisak*, 1981]. Furthermore, for this flow configuration, the process of mixing is partially reversible. Dipole tests with step input were evaluated by *Gelhar et al*. [1992] as R3 since the spreading is dominated by the flow field's nonuniformity and the BTCs are influenced by dispersion only in very early time [*Hoopes and Harleman*, 1967]. However, if sufficient early time data are available (and the analysis is performed according to the flow configuration), this kind of test is evaluated as R2. If the amount of early time data is sparse, the test configuration is not sensitive toward dispersivity and the studies are excluded [e.g., *Robson*, 1974].



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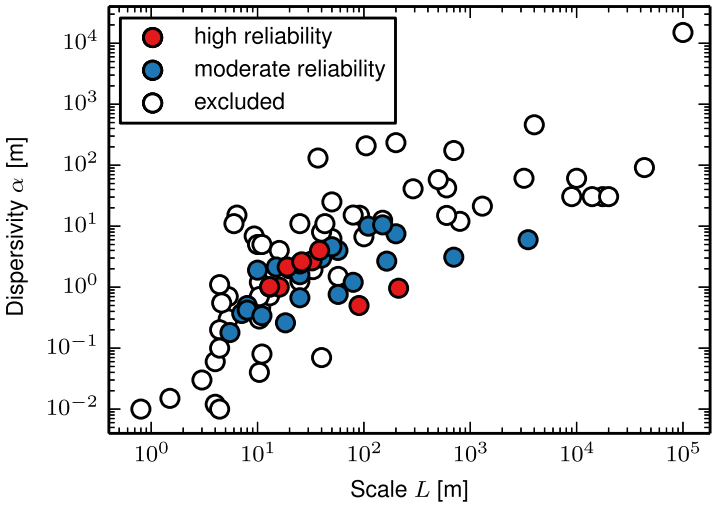


**Figure 2.** Macrodispersivities reported by *Gelhar et al*. [1992] with revised reliability. Values are listed in Tables 2 and 3.

Second, we specify exclusion criteria according to the quality of the data interpretation in combination with experimental conditions: (i) The nonconservative effects of the tracer (radioactive decay, adsorption, degradation, or chemical reactions) were neglected in the analysis [e.g., *Werner*, 1983]. This is particularly true for the apparent outlier of Figure 1, with a macrodispersivity value of order of $10^4$ m [*Wood*, 1981]. (ii) The BTC analysis is based on very few measurements in space and time; e.g., *Rousselot et al*. [1977] and *Sauty* [1977] reported a value which relied on 6 measurements in time, where 5 out of them only report the recovery phase. Tracer arrival and peak concentration are most probably missed by the measurements leading to an overestimation of the macrodispersivity. (iii) The measured BTC was outside the flow path, which resulted in missing the peak of the BTC [e.g., *Bierschenk*, 1959]. (iv) The analysis did not accord with the flow configuration; e.g., *Robson* [1974] used the method of *Grove and Beetem* [1971] in which a uniform flow solution is used to analyze a dipole test in highly nonuniform flow fields. (iv) The BTC was assumed to be a superposition of BTCs in separate hydro-stratigraphic units when their presence was not documented [e.g., *Rousselot et al*., 1977; *Sauty*, 1977]. (v) A bimodal curve was interpreted using a single peak method [e.g., *Hoehn and Santschi*, 1987].

Another reason for excluding a study is its focus on identification of $\alpha$ at small scales. According to *Pickens and Grisak* [1981] we have to distinguish between local dispersivities and macrodispersivity. Thus, in the case of a stratified formation, the former describes the dispersivity of an individual layer, whereas the latter is the dispersivity of the entire aquifer accounting for the effects of heterogeneity in conductivity. Since we focus on macrodispersivity, we refrain from including dispersivity values which are based on measurements at a particular depth [*Wood and Ehrlich*, 1978; *Lee et al*., 1980; *Leland and Hillel*, 1982; *Hoehn*, 1983; *Moltyaner and Killey*, 1988]. Similarly, we exclude studies which use the ADE for analyzing spreading which is due solely to local dispersion.

We specify several criteria for case studies of contamination events analyzed by large-scale numerical models. For these kind of studies, very low sensitivity of identified macrodispersivity is frequently reported [*Sudicky et al*., 2010; *Antonacci et al*., 2013], which results from the simultaneous estimation of dispersivity with other parameters (e.g. conductivity, porosity, retardation rates) and/or source conditions, which have a major impact on the concentration distribution. We encountered additional problems in numerical case studies including ambiguity in source definition, ignoring physico-chemical processes, problems with the grid scale, and missing or very coarse data calibration. Due to the aforementioned reasons, multiple studies were excluded [*Pinder*, 1973; *Robson*, 1974, 1978; *Konikow and Bredehoeft*, 1974; *Fried*, 1975; *Konikow*, 1976; *Helweg and Labadie*, 1977; *Ahlstrom et al*., 1977; *Vaccaro and Bolke*, 1983]. A detailed discussion is given in the supporting information.

The uncertainty in the source history is a general problem of contamination events. In studies using analytical solutions of the ADE instead of numerical models, the source quantity and temporal distribution of the contamination event are either estimated or another unknown in the equation. Since the concentration distribution is highly sensitive to the source history, the reliability of a dispersivity estimate under unknown source conditions is highly questionable [e.g., *Thorpe and Barry*, 1977; *Daniels*, 1982; *Egboka et al*., 1983].

However, not all contamination events have been excluded. In a few studies, the source was thoroughly reconstructed, given that the estimated dispersivity was able to be evaluated as moderately reliable [e.g., *Sykes et al*., 1982a; *Hoehn and Santschi*, 1987; *Naymik and Barcelona*, 1981].

For some studies, multiple interpretations of the same experiment by different authors exist. For example, an experiment in Palo Alto, California was interpreted by *Roberts et al*. [1981] and *Valocchi et al*. [1981]. We



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agree with the value of *Valocchi et al*. [1981], which was evaluated as R1, whereas the value of *Roberts et al*. [1981] was R3. *Gelhar et al*. [1992] reported alternative data interpretations for three experiments [*Wilson*, 1971; *Sauty*, 1977; *Sudicky et al*., 1983] performed by *Gelhar et al*. [1985] and *Welty and Gelhar* [1989], which we report as moderately reliable data instead of the antecedent interpretations.

The final result of the reevaluation of *Gelhar et al*. [1992] data is encapsulated by Figure 2, which displays the maintained macrodispersivity values of reliability classes R1 or R2, as well as the excluded ones. While the result will be discussed in the next section (section 3), it is worthwhile to mention here that the process resulted in a great reduction of the number of data and its scatter, particularly of large or very low macrodispersivity values.

### 2.4. Additional Dispersivity Data

Several new field studies, which focused on macrodispersivity, were conducted in the past 20 years; thus not covered by *Gelhar et al*. [1992]. We performed an exhaustive literature research to collect additional reliable dispersivity values which are represented in Figure 3. We reevaluated all studies according to the updated criteria of reliability given in Table 1. Here follows a brief discussion; details on the evaluation of reliability can be found in the supporting information.

*Schulze-Makuch* [2005] provided a review of recent studies on longitudinal macrodispersivities (see Introduction). Out of the 19 reported experiments in unconsolidated media, we reevaluated 3 as highly reliable (R1), 8 of moderate reliability (R2) and 8 were excluded (Table 2). Among the excluded, 4 studies are laboratory investigations (not relevant to the field of macrodispersivity), 3 were excluded due to quality reasons and 1 focused on local dispersivities determined at different depths of the aquifer. Two values of R2 result from a reanalysis of studies which originally focused on estimating local dispersivity. We determined the macrodispersivity from BTC analysis associated with the depth-averaged concentration. Details are provided in the supporting information.

Furthermore, we were able to identify 9 additional studies reporting dispersivity values, which we qualify as of moderate and high reliability. Values are listed in Table 2; detailed discussion is given in the supporting information. If an evolution of dispersivity with travel distance was given, we included all values (e.g., the Krauthausen experiment [*Vereecken et al*., 2000]). In a similar manner, we treated the highly reliable studies at Borden [*Freyberg*, 1986] and Cape Cod [*Garabedian et al*., 1991] for which *Gelhar et al*. [1992] reported the asymptotic value at the longest distance only. Values are summarized in Table S1 in the supporting information.

The summary of all newly added and remaining dispersivity values of reliability classes R1 or R2 is displayed in Figure 3. The figure includes dispersivities of multiple experiments at one field site and multiple values of one experiment for those in which a scale-dependent evolution of the dispersivity was reported. A discussion on the possible use of the data to explain the scale dependence of dispersivity is given in the next section. It is worthwhile to mention that all newly added data did not essentially change the picture, but rather supported the previously found range of reliable dispersivity values.

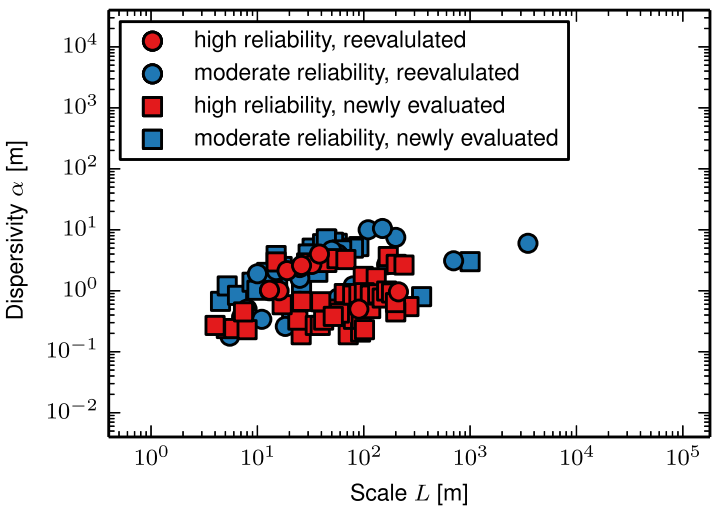


**Figure 3.** Reevaluated and newly evaluated macrodispersivities of high and moderate reliability, including multiple values of dispersivity from several field sites (Table 2 and supporting information Table S1).

## 3. Analysis of Scale Dependence of Macrodispersivity Data

After the thorough examination and screening of field data



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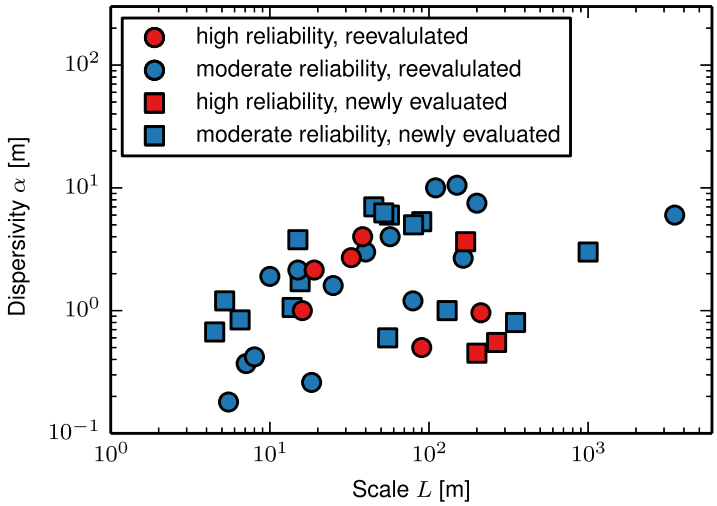


**Figure 4.** Reevaluated and newly evaluated macrodispersivities of high and moderate reliability, reduced to one value per field site and experiment, namely the value of highest reliability (Table 2).

reported in the literature, we proceed with the discussion on their possible use in order to reassess the macrodispersivity scale dependence concept. As a first step, we perform an aquifer-independent analysis by employing the raw data of Figure 3 for the ensemble of aquifers, but after an additional screening to eliminate possible bias. Next, we discuss the findings from a few thorough experiments in which macrodispersivity was determined as a function of distance. A computational example is given in the Appendix A3, where a similar shaped cloud of data points is created as in Figure 3 by theory. The interested reader may find that a collection of dispersivity measurements can be constructed without assuming transferability of aquifer parameters (Figure 6).

### 3.1. Aquifer-Independent Analysis

Several authors used the data from *Gelhar et al*. [1992] for a regression curve analysis, compiling data points for $\alpha$ as a function of $L$ from different formations (as discussed in the Introduction). However, the database of *Gelhar et al*. [1992] is not appropriate for a proper regression analysis relying on the ensemble of aquifers since it includes multiple values for some experiments and field sites, which creates a bias for those sites. For instance, 6 of the 12 values of high reliability (R1) in the data of *Gelhar et al*. [1992] (Figure 1) are from one experiment at Bonnaud, France [*Molinari and Peaudecerf*, 1977; *Sauty*, 1977]. And the total database includes six values from four experiments at the Borden site.

To create an unbiased representation, we reduced the data of Figure 3 by attaching only one value to each field site, namely the value of highest reliability at the longest distance. The result of this additional screening is represented in Figure 4, which displays an updated collection of reliable macrodispersivities from the literature in which every aquifer is represented by one value solely.

Inspection of Figure 4 reveals a scatter of data with macrodispersivities varying within two orders of magnitude over scales varying by four orders of magnitude. High and moderately reliable dispersivities do not differ significantly in the range of values observed in the field. Adding new data supports the results with regard to the range of observed value, but do not fundamentally change the picture of *Gelhar et al*. [1992] when limiting to data that are of high and moderate reliability (R1 and R2). Dispersivities at short distances ($<$ 10 m) are probably influenced by preasymptotic effects.

We conclude that the data ensemble of Figure 4 precludes performing a regression analysis toward a unique scale dependence. Indeed, a single dispersivity value for a certain distance, as predicted by a scaling law, cannot capture the complex distribution of dispersivities observed in different aquifers. For instance, at a distance $L$ between 125 and 150 m, we find macrodispersivities ranging from $\alpha$=0.45 m up to $\alpha$=10.5 m. Although Figure 4 shows an apparent increase of $\alpha$ with $L$, there are highly reliable macrodispersivities of low value at very long distances (for instance $\alpha$=0.55 m (R1) for $L = 266$ m at the Chalk River site [*Moltyaner et al*., 1993]); and high values for dispersivities at short distances (for instance $\alpha = 4$ m (R1) for $L = 38$ m at the Mobile site [*Huyakorn et al*., 1986a]). Hence, a particular dispersivity value for a certain distance as inferred by a scaling law is not able to predict those values for these field sites. Furthermore, the reliable dispersivities of Figure 4 do not support the conjectured unbounded growth of macrodispersivity with distance for unconsolidated aquifers, as implied by the various regression lines reviewed in the Introduction.



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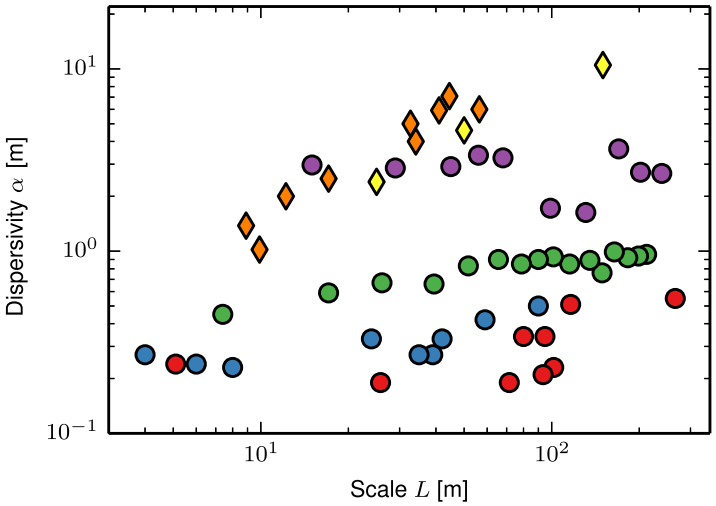


**Figure 5.** Macrodispersivity versus scale for a few selected aquifers: red—Chalk River [*Moltyaner et al.*, 1993], blue—Borden [*Freyberg*, 1986; *Rajaram and Gelhar*, 1991], green—Cape Cod [*Garabedian et al.*, 1991], purple—Krauthausen [*Vereecken et al.*, 2000], orange—Horkheimer Insel [*Ptak and Teutsch*, 1994], yellow—Corbas [*Sauty*, 1977; *Welty and Gelhar*, 1989]. Values are summarized in Table 1 in the supporting information.

### 3.2. Aquifer-Specific Analysis

In contrast with Figure 4, which provides unique representative macrodispersivity values for many aquifers, we concentrate now on a few specific aquifers for which thorough experiments have determined the change of $\alpha$ with $L$ (Figure 5). The references and asymptotic macrodispersivity values are displayed in Table 2; raw data are provided in supporting information Table S1. Of particular significance are the log-conductivity variance $\sigma_Y^2$ and the longitudinal correlation scale $\ell$ values. Most of these data were not available to *Gelhar et al.* [1992] since they are recent findings.

For all experiments in Figure 5, we observe a general trend of increasing dispersivity with distance. The studies of high reliability, in which a spatial moment analysis was performed, exhibit an asymptotic behavior. The sandy aquifers of low heterogeneity like at sites in Borden [*Freyberg*, 1986] ($\sigma^2=0.24$, $\ell=2.7$ m), Chalk River [*Moltyaner et al.*, 1993; *Dagan and Neuman*, 1997] ($\sigma^2=0.23$, $\ell=1.5$ m) and Cape Cod [*Garabedian et al.*, 1991] ($\sigma^2=0.24$, $\ell=2.6$ m), exhibit an asymptotic value in the range of 1 m. The value at the more heterogeneous Krauthausen site [*Vereecken et al.*, 2000] ($\sigma^2=1.08$, $\ell=6.7$ m) is in the order of 4 m. For the highly heterogeneous sites of Horkheimer Island [*Ptak and Teutsch*, 1994] ($\sigma^2=2.34$, $\ell=6-10$ m) and Corbas [*Sauty*, 1977], the travel distance of the tracer was apparently too short to arrive at constant asymptotic behavior.

Scale-dependent measurements of macrodispersivity from individual experiments reveal significant differences among aquifers due to their specific heterogeneity structure, a finding which is in agreement with the general theoretical analysis of Appendix A and in particular those of Appendix A3 and Figure 6. A unique scaling law cannot capture this behavior. Furthermore, an unbounded growth of macrodispersivity with travel distance is not supported by experimental data of sufficient reliability. The data displayed in Figure 5 strengthen the results of the stochastic theory regarding the growth of $\alpha$ with $L$ from the injection site toward an asymptotic, constant value. This conclusion is valid for aquifers of weak to moderate heterogeneity.

In contrast, the different behavior of highly heterogeneous aquifers displayed in Figure 5 is underscored by the findings of the tracer transport experiment, referred to as MADE-1, which was carried out at the Columbus Air Base aquifer in 1986 [e.g., *Boggs et al.*, 1992, *Adams and Gelhar*, 1992]. The unique features of this test, which distinguishes it from those referred to so far, includes the high degree of heterogeneity of the aquifer ($\sigma_Y^2 \cong 7$) and the occurrence of a highly skewed plume, markedly different from a Gaussian one. Different attempts to interpret the result were published over the last twenty years (for a review see *Zheng et al.* [2011]). The macrodispersivity $\alpha$ as defined by the second-order moment is not a meaningful parameter as it is highly dependent on the thin and prolonged tail of the BTC. Recently, *Fiori et al.* [2013] have applied a model of flow and transport based on the recent measurements of permeability spatial distribution of *Bohling et al.* [2012], which was able to predict satisfactorily the observed snapshots of $m(x, t)$. *Dogan et al.* [2014] demonstrated that the classical ADE with lab-scale dispersivity can successfully match the tracer behavior at MADE if the fine-scale variability of the hydraulic conductivity field is mapped out. These findings strengthen the conclusion of section 3.2 and the general ones of the Appendix A: transport is aquifer-specific and prediction requires characterization and modeling pertinent to the particular structure.



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## 4. Summary and Conclusion

In this paper we assess the factual base provided by field experiments to the notion of unique scaling of the longitudinal macrodispersivity $\alpha$. General theoretical considerations (Appendix A) indicate that for aquifers of stationary random permeability, mean uniform flow and ergodic plumes, $\alpha$ increases with $L$ and reaches an asymptotic constant value. Furthermore, for weakly and mildly heterogeneous aquifer formations an explicit dependence of $\alpha$ upon the log-conductivity statistical parameters pertinent to the specific aquifer structure and flow conditions was established [see e.g., *Dagan*, 1989; *Gelhar*, 1993; *Rubin*, 2003]. These theoretical findings are supported by extensive analysis of several field experiments (see Sect. 3.2).

However, based on macrodispersivity values reported from various field experiments over the years and of variable reliability [*Gelhar et al*., 1992], an alternative concept was conjectured. It states that $\alpha$ in unconsolidated material follows a unique scaling law of unbounded increase with $L$, which does not depend on the geological settings [e.g., *Neuman*, 1990]. Because of the apparent generality of the unique scaling approach, it is increasingly used in practice, where its appeal stems from the fact that field data are usually rather limited [*Fetter*, 2000], making it hard to derive site-specific models for $\alpha$. We note, however, that *Gelhar et al*. [1992] advised against a unique scaling approach and an unbounded increase of $\alpha$, and further, that the regression analyses used to establish the unique scaling models are characterized by a huge range, from as low as 0.1 m for travel distances shorter than one meter to more than 250 m for travel distances of the order of 10, 000 m. Thus, given the large number of studies covered, the limited number of reliable data in *Gelhar et al*. [1992], and the addition of many more studies conducted since then, it was found imperative to revisit the concept of unique scaling and to reevaluate its factual base.

We reanalyzed more than 50 studies cited by *Gelhar et al*. [1992], which report dispersivities from experiments in unconsolidated media. All dispersivity values were initially categorized by *Gelhar et al*. [1992] in three reliability classes. We expanded the classification criteria (Table 1) and reduced the data to moderately and highly reliable solely. The low reliability experiments were reevaluated and either excluded or upgraded. The exclusion led to a significant reduction of the number of data points. Consequently, the remaining 27 high and moderate reliability data for unconsolidated aquifers displayed a greatly reduced range of macrodispersivity values, between 0.1 m and 10 m, for travel distances up to 3, 500 m.

In addition to the reanalysis of *Gelhar et al*. [1992], we conducted an exhaustive literature research of field experiments conducted since the publication of *Gelhar et al*. [1992]. Experiments were analyzed and classified using the same criteria used for *Gelhar et al*. [1992] data, resulting in 20 additional high and moderate reliability macrodispersivity data. Altogether, 47 moderate-to-high quality data points were retained and analyzed in search for evidence to support the concept of unique scaling.

Remarkably, the complete data set with 47 high and moderate reliability field experiments displays a pattern similar to the one that emerged from the reanalysis of *Gelhar et al*. [1992]. (see Figure 2), which confirmed the consistency of our analysis and and the proposed classification. Examining the 27 points from *Gelhar et al*. [1992] and the expanded data set of 47 data points, neither provided support for the notion of unique scaling of $\alpha$, nor a clear tendency of increase with $L$. Thus, in absence of field evidence to support the concept of a unique scaling in macrodispersivity, its adoption for data reduction (by regression) and for modeling $\alpha$ appears to be presently unjustified.

Further evidence is provided by focusing on high and moderate reliability experiments, where extensive monitoring provided longitudinal dispersivity data for various travel distances. As shown in Figure 5 we could illustrate that at four weak and mildly heterogeneous sites (Borden, Cape Cod, Chalk River, Krauthausen), constant asymptotic macrodispersivity values were reached within a travel distance of less than 50 m. This is in agreement with the theoretical models presented in the literature and recalled in Appendix A. At the two more heterogeneous sites (Corbas and Horkheimer Insel) asymptotic values were not reached up to a travel distances of 50 m and 100 m, respectively, covered by the monitored plumes.

In the absence of a scaling law, adequate characterization of the hydrogeological conditions together with a careful delineation of the plume initial conditions seems to be a required precondition for predicting solute spreading. Site specific investigations are time consuming and also costly, but appear to be unavoidable at present. There is room, therefore, for further development of efficient characterization techniques, as well as advanced models, to analyze and predict transport by groundwater.

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## Appendix A: Macrodispersivity From Subsurface Transport Theory

### A1. Theoretical Foundation of Macrodispersivity

A few classical results about macrodispersivity are reviewed. Several equations and solutions are common tools for the analysis of field data. The theory is developed in a Lagrangian picture using a stochastic approach as given e.g., in *Dagan* [1989]. The aquifer's hydraulic conductivity is modeled as log-normal distributed spatial random function $K(\vec{x})$. Thus, the steady velocity field $U(\vec{x})$, as solution of the flow problem, is also random and characterized statistically by its various moments.

With regard to the transport problem, the concentration of an inert solute $C$ injected instantaneously into the volume $V$ is given by

$$C(\vec{x},t)=\int_V C_0(\vec{x}_0)\delta[\vec{x}-\vec{X}(t,\vec{x}_0)]\,\mathrm{d}\vec{x}_0 \tag{A1}$$

where $C_0$ is the initial concentration at $\vec{x}_0$ within $V$, $\delta$ is the Dirac operator and $\vec{x}=\vec{X}(t,\vec{x}_0)$ is the equation of the trajectory of a solute particle originating at $\vec{x}_0$.

Equation (A1) stipulates that an elementary mass of solute moves along the trajectory. The trajectory $\vec{X}$ is related to the velocity field by $\mathrm{d}\vec{X}/\mathrm{d}t=\vec{W}(t,\vec{x}_0)+\vec{w}_d$ where $\vec{W}$ is the Lagrangian velocity of a fluid particle originating at $\vec{x}_0$ while $\vec{w}_d$ is velocity of a Brownian motion process, modeling pore-scale dispersion.

The rate of spreading of the plume is quantified by its first and second spatial moment, defined with the aid of (A1) by

$$\vec{L}(t)=\frac{1}{V}\int_V \vec{X}(t,\vec{x}_0)\,\mathrm{d}\vec{x}_0 \tag{A2}$$

$$S_{ij}(t)=\frac{1}{V}\int_V (X_i-L_i)(X_j-L_j)\,\mathrm{d}\vec{x}_0 \tag{A3}$$

The problem is further simplified by assuming that the mean velocity $\vec{U}=\langle\vec{W}\rangle$ is constant and aligned with the $x_1$ axis. Moreover, it is assumed that the plume is sufficiently large to justify ergodicity of the moments. The cases in which the plume size is not large with respect to the heterogeneity scale were analyzed in the literature by using the concept of relative dispersion with respect to the centroid [see e.g., *Rajaram and Gelhar*, 1995; *Rubin et al.*, 1999; *Lawrence and Rubin*, 2007]. Thus, concentrating here on the longitudinal moments, one gets $L_1(t)\simeq Ut$ and $S_{11}(t)\simeq\langle(X_1-Ut)^2\rangle=X_{11}(t)$, where $X_{11}$ is the variance of the fluctuation of the longitudinal component $X_1-Ut$ of the trajectory. It is assumed that the velocity and trajectory fields are stationary so that $X_{11}$ does not depend on the initial position $\vec{x}_0$. It can be calculated from the variance of the longitudinal Lagrangian fluid velocity fluctuation $w_{11}$ as $X_{11}=2\int_0^t(t-t')\;w_{11}(t')\,\mathrm{d}t'$.

The longitudinal macrodispersivity is defined by $\alpha=\frac{1}{2U}\frac{\mathrm{d}S_{11}}{\mathrm{d}t}$, this choice being motivated by the fact that for a diffusive Gaussian plume this is precisely the relationship between the diffusivity and the second spatial moment. However, this definition is adopted in the groundwater literature even if the concentration plume is not Gaussian. Furthermore, it served for identifying $\alpha$ from concentration measurements in a few field experiments (see section 2).

Following (A3) and the specification of $X_{11}$ one obtains the basic Taylor relationship [*Taylor*, 1921]

$$\alpha(t)U=\frac{1}{2}\frac{\mathrm{d}S_{11}}{\mathrm{d}t}=\frac{1}{2}\frac{\mathrm{d}X_{11}}{\mathrm{d}t}=\int_0^t w_{11}(t')\,\mathrm{d}t'=\int_0^t \sigma_w^2\rho_w(t')\,\mathrm{d}t' \tag{A4}$$

where $\sigma_w^2$ and $\rho_w(t)$ are the Lagrangian velocity variance and auto-correlation, respectively.

As an alternative to the plume spatial distribution, a common measure of transport is by mass arrival at fixed control planes at $\vec{x}$ by breakthrough curve (BTC) analysis. The latter, representing the probability density function (pdf) of solute particles travel time $\tau$, can be analyzed in a manner similar to the spatial distribution [see e.g., *Dagan et al.*, 1992]. In the Fickian stage, for which the pdf of $\tau$ is inverse Gaussian, such an analysis leads to

$$\alpha = (U/2)\, \mathrm{d}\sigma_\tau^2/\mathrm{d}t \tag{A5}$$

where $\sigma_\tau^2$ is the travel time variance or the BTC second temporal moment. This relationship was used in field experiments in order to identify the macrodispersivity, even if the plume was not Gaussian.

A central issue of stochastic modeling of groundwater flow and transport is to derive the dependence of the velocity and trajectories fields upon the permeability $K(\vec{x})$ and the flow conditions. A central result was derived for aquifers of weak heterogeneity, at first order in the log-conductivity variance $\sigma_Y^2$ [*Dagan*, 1989]. The asymptotic macrodispersivity is given by $\alpha = \sigma_Y^2\ \ell$, where $\ell$ is log-conductivity integral scale in the mean flow direction.

### A2. Single Aquifer Scale-Dependence of Macrodispersivity

Equation (A4) allows to analyze a few important general properties of $\alpha$. Two limits of $\alpha$ are of particular significance: (*i*) the limit for small times $\alpha U \to \sigma_w^2\ t$ for $t \to 0$; and (*ii*) the limit for large times $\alpha U \to \sigma_w^2 T_w$ for $t \to \infty$, where $T_w = \int_0^\infty \rho_w(t)\, \mathrm{d}t$ is the integral scale of the Lagrangian velocity, also known also as the macro scale.

In the early time limit $\alpha \sim \sigma_w^2 t/U$ grows linearly with time or with the distance covered by the centroid $L = Ut$ as $\alpha \sim \sigma_w^2 L$. Thus $\alpha$ is scale dependent in the short time limit. However, the slope is dependent on the particular aquifer structure, as $\sigma_w^2$ is related to the hydraulic conductivity through the flow equations.

In the long time limit and for a Lagrangian velocity of finite integral scale $T_w$, $\alpha$ tends to a constant, and transport becomes Fickian. A Fickian regime is approximately reached for a finite, but large, time $t \gg T_w$ and by the central limit theory. The point of is that $\alpha$ depends on two independent parameters, which in turn depend on the heterogeneous structure: $\sigma_w^2$ related to the degree of aquifer heterogeneity and $T_w$ related to the correlation structure.

At any intermediate time $\alpha(t)U$ as given in equation (A4) is scale dependent. However, its rate of growth with scale $L = Ut$ depends on the particular shape of the auto-correlation $\rho_w(t)$. For such a time period transport may seem to be anomalous, which is a topic of wide debate in the literature. In particular, anomalous-like behavior is encountered in highly heterogeneous formations, for which $T_w$ may be large relative to the travel time of the plume in applications [see e.g., *Fiori et al.*, 2007].

Based on equation (A4) it can be seen, that there is no theoretical foundation to the assumption that $\alpha$ has a unique dependence upon $L = Ut$. It is specific to aquifer structure and flow conditions as reflected by the statistical moments of the Lagrangian velocity field or of the travel time distribution. Matters are even more complex for nonergodic plumes, for which dispersivity may also depend on the plume size and its history.

### A3. Computational Example

The idea of a unique scaling law as derived from a collection of field data is based on a the idea of transferability of aquifer parameters. As discussed in section A2, macrodispersivity between aquifers is transferable if aquifers have the same statistical parameters. However, a collection of dispersivity measurements can result in a similar shaped cloud of data points as in Figures 3 and 4 without assuming transferability. We are going to show that in the following from the theoretical point by a simple computational example.

Following the lines of Sec. A.1, an explicit expression for dispersivity can be derived from equation (A4) for a first order approximation [*Dagan*, 1989]. The aquifer-specific scaling of macrodispersivity for pure advection is given by

$$\alpha(L, \sigma_Y^2, \ell) = \sigma_Y^2 \ell^2 \left( \frac{1}{\ell} - \frac{4\ell}{L^2} \left( \frac{2\mathrm{e}^{-L/\ell}\ell^2}{L^2} - \frac{6\ell^2}{L^2} + \frac{4\mathrm{e}^{-L/\ell}\ell(1+\ell/L)}{L} + 2\mathrm{e}^{-L/\ell}(1+\ell/L) + 1 \right) \right) \tag{A6}$$

with $L$ being the distance from the injection source, $\sigma_Y^2$ the log-conductivity variance and $\ell$ the correlation scale assuming a Gaussian correlation structure. Clearly, this expression is an approximation, since the issue of ergodicity, effects of pore-scale dispersion and log conductivity statistics are typically different than assumed at first order. Nevertheless the first order expression has shown to be rather robust.

For designing a random collection of field data, we take 5 measurements at random locations in the range of $L \in [5\ \mathrm{m};\ 300\ \mathrm{m}]$ from 50 random aquifers. The log-conductivity statistics of the aquifers are uniformly distributed with $\sigma_Y^2 \in [0.1; 1]$ and $\ell \in [0.2\ \mathrm{m}; 10\ \mathrm{m}]$. For simplicity we assume uniform distribution of values. The ranges of $L$ and $\ell$ are chosen similar to those of the aquifers in Figure 5. The variance $\sigma_Y^2$ is limited to 1



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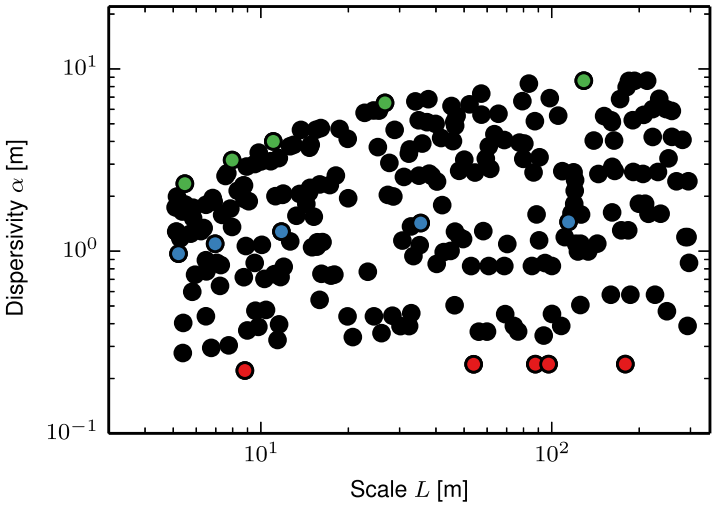


**Figure 6.** Macrodispersivity versus scale for 50 synthetic aquifers of random variance $\sigma_Y^2 \in [0.1; 1]$ and correlation scale $\ell \in [0.2$ m;10 m]. Dispersivities were calculated for every aquifer at 5 random locations $L \in [5$ m;300 m] according to equation (A6). Coloring (red, blue and green) indicate the dispersivities versus scale for three selected aquifers.

because of the first order approximation. The number of 50 aquifers is chosen according to the number of experiments, which are finally evaluated as moderately or highly reliable. We calculate the macrodispersivity according to $\alpha$ in equation (A6) and we get a data cloud as given in Figure 6.

Qualitatively, Figure 6 show similar behavior as Figure 5. Data points cluster on the right and we observe a kind of asymptotic increase of the cloud. These findings are in line with the construction of Figure 6, containing values of aquifers with different statistical parameters similar to those in as Figure 5.

Comparing Figure 6 to the aquifer independent collection of field data (Figures 3 and 4) reveals significant similarities. Except for short distances values of macrodispersivities are encountered at all combinations of $\alpha$ and $L$. Thus, a plot of dispersivity field data $\alpha$ versus scale $L$ does not allow to infer on transferability of aquifer parameters.

Taken all together the data in Figure 6 are pretty much randomized as they come from different aquifer settings, including cases with same or close variance $\sigma_Y^2$ and correlation scale $\ell$ but different $L$. As based on $\alpha$ (equation (A6)), macrodispersivities exhibit a scale dependency, but this features is aquifer specific. A collection of values as given in Figure 6, or similarly the field data in Figures 3 and 4, cannot serve as foundation for a unique scaling law, which provides macrodispersivities based on the distance $L$, but independent of the aquifer specific properties.

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The data used to produce the results of this paper are taken from literature, with references to all used publications. A summary of the data are given in Tables 2 and 3. A detailed discussion on the individual studies including the reliability of the dispersivity data is given in as supporting information. We thank the Helmholtz Centre for Environmental Research—UFZ for financial support of this work. We are grateful to L. Jakobs and F. Heße for improving the English phrasing of the manuscript. The constructive comments of three anonymous reviewers are greatly appreciated.

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